

PROFESSIONAL SUMMARY

Robotics and autonomous systems engineer with experience in ROS/ROS 2, computer vision, sensor fusion, localization, path planning, control, and mobile robotics research. Strong mechanical engineering foundation in dynamics, numerical modeling, sensing, signal processing, system identification, experimental validation, and advanced manufacturing. Research experience spans autonomous mobile robots at the University of Waterloo and vibration-based system identification at the University of Victoria. Currently developing and validating ROS 2-based robotics systems and perception pipelines with an emphasis on quantitative evaluation and reproducible implementation.

EDUCATION

University of Waterloo	2020–2021
PhD Program in Mechanical & Mechatronics Engineering	
Voluntary Withdrawal	
Graduate Funding Package: CAD 100,000 guaranteed upon admission	
University of Victoria	2016–2019
Master of Applied Science (MASC), Mechanical Engineering	
Thesis: Predicting Regenerative Chatter in Turning Using Operational Modal Analysis	
Pusan National University	2005–2012
Bachelor of Science (B.Sc.), Mechanical Engineering	
Undergraduate Thesis: Investigation of SOFC Injecting Natural Gas Directly	

SELECTED ROBOTICS PROJECTS

Real-Time Vision Perception Pipeline

Personal Robotics Project

ROS 2 · PyTorch · Ultralytics YOLO · OpenCV · DeepSORT

- Developed a ROS 2 perception pipeline integrating YOLO-based instance segmentation, multi-object tracking, image processing, and quantitative performance logging.
- Achieved 31.95 FPS across 2,136 steady-state frames, with 18.26 ms average processing latency and 24.87 ms P95 latency on an NVIDIA RTX 5060 Laptop GPU.
- Measured 8.70 ms average inference latency and 12.52 ms P95 inference latency, processing scenes containing up to 11 detected objects per frame.
- Structured the implementation as reusable ROS 2 nodes with launch configuration, CSV-based performance reporting, Git version control, and documented experimental results.

ROS 2 EKF Sensor Fusion

Personal Robotics Project

ROS 2 Humble · Gazebo

- Implemented an Extended Kalman Filter pipeline for mobile-robot state estimation using wheel odometry and IMU measurements.
- Integrated simulated sensor streams, coordinate frames, and estimated robot states within a Gazebo-based ROS 2 environment.
- Evaluated fused-state behavior against individual sensor measurements and documented the implementation as a reproducible robotics demonstration.

LiDAR-Camera Fusion for Road Segmentation

University of Waterloo

KITTI · TensorFlow/Keras · Computer Vision

- Developed and evaluated RGB and LiDAR-camera fusion architectures for semantic road segmentation using projected Velodyne depth data and the KITTI road dataset.
- Compared RGB-only, early-fusion, and late-fusion variants of a modified SegNet architecture.
- Achieved a MaxF score of 0.9364 using late fusion, compared with 0.9010 for the RGB-only model, and evaluated accuracy, IoU, computational cost, and fusion-strategy trade-offs.

RESEARCH EXPERIENCE

University of Waterloo

May 2020 – Dec 2021

Research Assistant, Mechanical & Mechatronics Engineering

Waterloo Mechanical Systems and Control Lab

- Conducted mobile robotics research involving autonomous navigation, path planning, localization, sensing, and physical-robot experimentation.
- Investigated transfer-learning approaches for improving learning efficiency in robot semantic tasks.
- Developed LiDAR-camera fusion methods for semantic road segmentation using deep neural networks and the KITTI dataset.
- Implemented model-predictive-control-based mobile-robot navigation and conducted physical experiments to evaluate path-following performance.
- Supported research on path-planning optimization for indoor mobile robots operating in dynamic environments.
- Conducted physical experiments using Summit mobile robots, motion-capture systems, sensors, and associated robotic hardware.
- Supported physical validation of motion-aware navigation and planning approaches.
- Maintained robotic systems, sensors, accessories, and experimental platforms used for laboratory research.
- Developed robot-operation documentation and technical instructional material.

University of Victoria

Aug 2017 – Apr 2019

Research Assistant, Dynamics and Digital Manufacturing Lab

Operational Modal Analysis for Chatter Prediction in Turning

- Developed an Operational Modal Analysis approach for estimating machining stability from in-process vibration measurements before regenerative chatter fully developed.
- Modeled turning dynamics using multi-degree-of-freedom, state-space, and time-delay formulations.
- Performed numerical simulation, pole identification, modal damping estimation, and stability analysis in MATLAB.
- Designed and conducted instrumented machining experiments using accelerometers, National Instruments data-acquisition hardware, acoustic measurements, and experimental modal analysis.
- Applied correlation analysis, power spectral density analysis, Time Domain Poly Reference identification, and modified Least Squares Complex Exponential methods.
- Investigated separation of structural modes from deterministic spindle harmonics in operational vibration measurements.
- Conducted finite-element and modal analyses related to machining-system structural dynamics.
- Validated analytical and numerical stability predictions against physical machining experiments.

Micromilling Dynamics & Laser Doppler Vibrometry

- Conducted non-contact vibration measurements of a high-speed micromilling system using a Laser Doppler Vibrometer.
- Used broadband acoustic excitation to characterize tool-tip structural dynamics.
- Applied Operational Modal Analysis to estimate modal frequencies and damping ratios from output-only vibration measurements.
- Compared OMA results with conventional impact-hammer Experimental Modal Analysis.
- Investigated spindle-harmonic interference at multiple rotational speeds and applied modified LSCE methods to distinguish harmonic excitation from structural modes.
- Performed MATLAB-based signal processing using correlation functions, power spectral density, modal identification, and stabilization analysis.

University of Victoria

Jun 2016 – Jul 2017

Research Assistant, Laboratory of Advanced Multi-scale Manufacturing

- Fabricated and tested fiber-optic sensors, including Long Period Grating (LPG) and Fiber Bragg Grating (FBG) devices.
- Conducted femtosecond-laser machining of glass and other materials.
- Fabricated microfluidic channels in glass using femtosecond-laser processing.
- Investigated glass welding for microfluidic devices without intermediate bonding layers.
- Conducted nanoparticle coating experiments on glass substrates.
- Supported experimental setup, fabrication, testing, and engineering analysis.

PROFESSIONAL EXPERIENCE

MKS Investments

2023–2026

Manager

- Reviewed and validated technical and market analyses produced by internal analysts and AI-assisted workflows, identifying incorrect assumptions, data inconsistencies, unsupported claims, and source-quality issues.
- Cross-checked technical feasibility, benchmarks, source data, and market evidence to improve the reliability of technology-development and investment decisions.
- Managed technology and market research across emerging industries, synthesizing technical, commercial, and competitive information.
- Evaluated technologies and translated technical findings into decision-oriented recommendations for management and external stakeholders.
- Supported business-development activities through technology evaluation, research synthesis, and market analysis.

Kyungdong LNG

2010

Co-op Student

- Monitored operating conditions of LNG storage tanks and LNG pressure-reduction facilities.
- Supported routine operational monitoring and equipment-status checks.

TEACHING EXPERIENCE

University of Waterloo

Autonomous Mobile Robotics — MTE 544

Teaching Assistant | Fall 2021

- Customized laboratory manuals for students in the Mechatronics Engineering program.
- Provided instruction and technical support for programming in the Robot Operating System (ROS) environment.
- Supported laboratory exercises involving ROS, SLAM, localization, navigation, and mobile robotics.
- Evaluated assignments and examinations and provided technical feedback.

University of Waterloo

Autonomous Mobile Robotics — ME 597c

Teaching Assistant | Winter 2021

- Developed and modernized laboratory manuals for remote robotics instruction.
- Supervised laboratory activities and evaluated laboratory reports.
- Provided instruction and troubleshooting support for ROS-based mobile robotics.
- Created demonstration videos illustrating mobile-robot maneuvering and navigation behavior.

Robotics Laboratory Development

- Prepared and validated ROS, Gazebo, MATLAB, and TurtleBot environments for remote laboratory exercises.
- Developed and validated laboratory material covering ROS fundamentals, SLAM, AMCL localization, and autonomous navigation.
- Developed AMCL parameter-study exercises examining particle count, measurement noise, resampling behavior, and computational trade-offs.
- Developed sampling-based path-planning exercises involving RRT and RRT*, robot geometry, obstacle inflation, collision validation, Reeds-Shepp and Dubins state spaces, and Pure Pursuit tracking.
- Supported live instruction, debugging, experimental verification, grading, and laboratory-report assessment.

University of Victoria

Fall 2017

Introduction to Mechanical Vibrations — MECH 330

Teaching Assistant

- Supported students with mechanical-vibration theory, analytical problem solving, and course material.
- Evaluated assignments, quizzes, and tests.
- Provided academic feedback and supervised examinations.

University of Victoria

Spring 2018

Mechanics of Solids II — MECH 320

Teaching Assistant

- Managed and supervised laboratory activities.
- Guided students through experimental procedures and engineering analysis.
- Maintained laboratory safety and proper experimental practice.
- Evaluated laboratory reports and assignments.

Teaching Assistant

- Managed laboratory operations and scheduling.
- Updated laboratory manuals and manufacturing-workflow documentation.
- Evaluated student project designs and supervised CNC machining activities.
- Provided instruction and guidance in CAD/CAM software and manufacturing workflows.
- Reviewed student designs, toolpaths, machining programs, coordinate systems, and setup details before physical machining.
- Supported SolidWorks, MasterCAM, VERICUT, G-code verification, and Haas 3-axis CNC machining.
- Assisted students from CAD design through CAM programming, machining verification, and final part production.
- Evaluated laboratory reports and provided technical feedback.

UNDERGRADUATE RESEARCH & ENGINEERING

Solid Oxide Fuel Cell System Fueled Directly by Natural Gas Pusan National University

2011–2012

- Developed and evaluated a Solid Oxide Fuel Cell (SOFC) system designed to utilize natural gas directly without an external reformer.
- Conducted fuel-cell system setup, experimental testing, and performance analysis.
- Investigated direct natural-gas utilization in tubular-type solid oxide fuel cells.
- Developed the research into a Korean patent application.
- Received university and external awards associated with the project.

Undergraduate Thesis: Investigation of SOFC Injecting Natural Gas Directly

SELECTED PUBLICATION

S. Kim and K. Ahmadi, "Estimation of vibration stability in turning using operational modal analysis," Mechanical Systems and Signal Processing, 2019.

PATENT

Solid Oxide Fuel Cell System Fueled By Natural Gas Directly Korean Patent Application No. 10-2012-0109237

2012

Inventor: Sooyong Kim et al.

SCHOLARSHIPS & AWARDS

- University of Waterloo Graduate Funding — CAD 40,000 | 2020–2021
- University of Victoria Scholarship — CAD 27,000 | 2016–2019
- Pusan National University Scholarship — CAD 10,000 | 2008–2011
- Grand Prize, Integrated Capstone Design Competition | 2011
- Grand Prize, Research Assessment of Industry-University Collaboration | 2011
- Excellence Award, Engineering Students Technology & Science Competition | 2011

TEACHING CERTIFICATIONS

Certificate in University Teaching Center for Teaching Excellence, University of Waterloo

Sep 2020

Certificate in TA Training Division of Learning and Teaching Support and Innovation, University of Victoria

Sep 2017

CORE TECHNOLOGIES

Robotics & Autonomy: ROS 2, ROS, Nav2, SLAM, AMCL, Localization, Path Planning, RRT/RRT*, Pure Pursuit, Model Predictive Control

Perception & Sensor Fusion: OpenCV, YOLO, Object Tracking, Instance Segmentation, Semantic Segmentation, LiDAR-Camera Fusion, EKF, IMU, Odometry

Programming & Computing: Python, C++, MATLAB, Linux, Git, GitHub, Jupyter

Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, Ultralytics YOLO, DeepSORT, Semantic Segmentation, Transfer Learning

Dynamics & System Analysis: State-Space Modeling, Time-Delay Systems, Numerical Analysis, System Identification, Operational Modal Analysis, Experimental Modal Analysis, Stability Analysis, Signal Processing, PSD, Correlation Analysis, FEA

Simulation & Robotics Platforms: Gazebo, TurtleBot, Summit XL

Experimental & Measurement Systems: Accelerometers, National Instruments DAQ, Laser Doppler Vibrometry, Acoustic Measurement, Motion Capture

Mechanical & Manufacturing Tools: SolidWorks, MasterCAM, VERICUT, CNC Machining, Femtosecond-Laser Processing